Answer Key

Mahatma Gandhi University, Kottayam

Fifth Semester -October 2026 Examination

MG5DSCFSQ301-DAIRY TECHNOLOGY

Part A – MCQ

Q. No.	Correct answer	
1	d	6.60%
2	a	Aseptic packaging
3	a	To dissolve proteins and liberate fat
4	a	Special milk
5	b	Appropriate separation of raw and heat-treated products
6	b	Save heating and cooling energy
7	c	Partial removal of whey from fermented curd
8	b	Clean flavour and smooth texture
9	b	Refrigeration
10	a	Ensures probiotic effectiveness

Part B – Short Essay Answers

11. Comment on the freezing point and boiling point of milk.

Freezing point of milk

- The normal freezing point of milk is approximately \(-0.54^\circ C\).
- It is lower than that of pure water because milk contains dissolved substances such as lactose and mineral salts.
- The freezing point is relatively constant in normal milk.
- Addition of water raises the freezing point towards \(0^\circ C\), making freezing-point measurement useful for detecting possible adulteration with water.

Boiling point of milk

- Milk boils at a temperature slightly above that of pure water at the same atmospheric pressure.
- At normal atmospheric pressure, its boiling point is approximately \(100.1–100.2^\circ C\).
- Dissolved solids in milk cause this slight elevation in boiling point.

- The boiling point decreases at lower atmospheric pressure and increases at higher pressure.

12. Describe the carbohydrate and mineral value of milk.

Carbohydrate value

- Lactose is the principal carbohydrate in milk.
- It is generally present at about 4.5–5.0%.
- Lactose provides energy and contributes to the mild sweetness of milk.
- It promotes calcium absorption and serves as a substrate for lactic acid bacteria during fermentation.
- Fermentation of lactose produces lactic acid, which is important in the manufacture of curd, yoghurt and cheese.

Mineral value

- Milk is a good source of calcium and phosphorus, which are essential for the development and maintenance of bones and teeth.
- Potassium helps maintain fluid balance and supports normal muscle and nerve function.
- Sodium and chloride help maintain electrolyte balance.
- Magnesium supports normal enzyme and muscle functions.
- Milk also contains small amounts of trace minerals, including zinc, iron and copper.

13. List and comment on the major types of cleaning compounds used in dairy industries.

1. Alkaline cleaners: Sodium hydroxide and related alkaline compounds remove milk fat, proteins and other organic residues.
2. Acid cleaners: Nitric acid and phosphoric acid remove mineral deposits, milkstone and inorganic residues.
3. Chlorinated cleaners: Chlorine-containing alkaline formulations help remove protein deposits and improve cleaning efficiency.
4. Surfactants: These reduce surface tension, improve wetting and help loosen and remove dirt from equipment surfaces.
5. Sequestering agents: Compounds such as phosphates and certain chelating agents bind calcium and magnesium ions, reducing mineral deposition.
6. Sanitizers: Chlorine compounds, peracetic acid and other approved sanitizers reduce microbial contamination after cleaning.

Note: Cleaning removes soil, whereas sanitization reduces microorganisms to acceptable levels. Correct concentration, temperature, contact time and rinsing are essential.

14. Explain the steps involved in the manufacture of cottage cheese.

Cottage cheese is a fresh cheese prepared by coagulating milk and separating the curd from whey.

Manufacturing steps:

1. Selection of milk: Good-quality milk is selected and filtered.
2. Standardization: Milk fat is adjusted according to the desired product.
3. Pasteurization: Milk is heat-treated to destroy pathogenic microorganisms and reduce spoilage organisms.
4. Cooling and inoculation: Milk is cooled to a suitable temperature and a starter culture is added.
5. Coagulation: The milk is allowed to develop acidity and form a curd. Rennet may also be used depending on the method.
6. Cutting the curd: The curd is cut into small cubes to facilitate whey separation.
7. Cooking: The curd is gradually heated to develop the required firmness and promote whey expulsion.
8. Draining: Whey is separated from the curd.
9. Washing and cooling: The curd may be washed with cold water to remove residual lactose and acid and to cool it.
10. Salting and dressing: Salt and cream dressing may be added, depending on the product type.
11. Packaging and storage: The cheese is packed hygienically and stored under refrigeration.

15. Explain the steps of churning and working in butter manufacturing.

A. Churning

1. Pasteurized cream is cooled and ripened when required.
2. The cream is brought to a suitable churning temperature.
3. It is agitated in a butter churn.
4. Agitation destabilizes the fat globule membrane and causes fat globules to aggregate.
5. Butter granules form and separate from the liquid buttermilk.
6. The buttermilk is drained off.

B. Working

1. The butter granules are worked mechanically to combine them into a uniform mass.
2. Moisture droplets are distributed throughout the butter.
3. Salt may be added according to the desired product.
4. Working controls the texture, consistency and moisture distribution.
5. Excessive working may produce an undesirable greasy or soft texture.
6. The finished butter is shaped, packaged and stored under refrigeration.

16. Discuss Whole Milk Powder (WMP).

Definition: Whole Milk Powder is a dairy product obtained by removing most of the water from milk while retaining its major milk constituents, including fat, protein, lactose and minerals.

Manufacturing process

1. Reception and filtration: Good-quality milk is received and filtered.
2. Standardization: Milk composition is adjusted to meet product specifications.
3. Pasteurization: Milk is heat-treated to improve microbiological safety.
4. Concentration: Water is removed under vacuum using an evaporator.
5. Homogenization: The concentrated milk may be homogenized to improve fat distribution.
6. Spray drying: Concentrated milk is atomized into hot air, rapidly removing moisture.
7. Cooling: The powder is cooled under controlled conditions.
8. Packaging: It is packed in moisture-resistant, airtight packaging to protect quality.

Composition and characteristics

- Contains milk fat, proteins, lactose, minerals and small amounts of moisture.
- Has a longer shelf life than liquid milk when properly processed and packaged.
- Should have a characteristic creamy colour, pleasant flavour and good reconstitution properties.

Uses

- Reconstitution into liquid milk.
- Manufacture of confectionery, bakery products, ice cream and chocolate.
- Preparation of beverages and other formulated food products.

Storage: WMP should be protected from moisture, oxygen, excessive heat and strong odours to prevent caking, oxidation and flavour deterioration.

Part C – Essay Answers

17. Explain the various factors affecting the transportation of raw milk in India.

Milk is highly perishable because it contains nutrients and moisture that support microbial growth. Proper transportation from farms and collection centres to dairy plants is essential to maintain milk quality and safety.

Factors affecting the transportation of raw milk

1. Distance from farm to dairy: Longer distances increase transportation time and the risk of microbial multiplication, particularly when milk is not adequately cooled.

2. Ambient temperature: High temperatures, common in many parts of India, accelerate bacterial growth and spoilage.

3. Cooling facilities: Availability of bulk milk coolers and refrigerated transportation helps maintain low temperatures and preserve quality.

4. Transportation time: Delays during collection, loading, transit and unloading increase the risk of deterioration.

5. Road and infrastructure conditions: Poor roads, traffic congestion and inadequate rural transport facilities may cause delays and physical damage to containers.

6. Type of transportation: Milk may be transported in cans, insulated vehicles or bulk milk tankers. The choice depends on the volume of milk, distance and available infrastructure.

7. Hygiene of containers and vehicles: Unclean cans, tankers and handling equipment may introduce microorganisms and contaminate milk.

8. Seasonal variations: Summer heat increases cooling requirements, while heavy rainfall and floods may disrupt transportation routes.

9. Milk collection system: Efficient village collection centres, cooperative networks and organized procurement systems help reduce delays and improve milk handling.

10. Handling practices: Rough handling, improper loading and unloading, and repeated exposure to the environment may affect milk quality.

11. Quality of raw milk: Initial microbial load, acidity, adulteration and the health of the milch animals influence the quality of milk during transportation.

12. Availability of trained personnel: Properly trained milk handlers and drivers help maintain hygiene, temperature control and accurate records.

Measures to improve transportation

- Cool milk promptly after milking.
- Use clean, food-grade milk cans and properly sanitized tankers.
- Maintain the cold chain and minimize transportation delays.
- Establish efficient collection centres and bulk milk cooling facilities.
- Conduct regular quality testing at collection and reception points.

Conclusion: Efficient transportation requires proper cooling, hygienic handling, suitable infrastructure and effective milk collection systems. These measures help preserve milk quality from the farm to the dairy plant.

18. Describe the procedure followed for unloading milk received in cans and tankers at the dairy reception dock.

Milk reception is the initial stage of dairy processing. It involves checking the quality and quantity of incoming milk before accepting it for processing.

A. Unloading milk received in cans

1. Reception and identification: Record the supplier's details, vehicle number, collection centre and time of arrival.
2. Inspection of cans: Examine cans for cleanliness, damage, leakage and proper closure.
3. Sampling: Collect representative samples from the cans using clean, sanitized sampling equipment.
4. Quality testing: Check temperature, appearance, smell, acidity and other required parameters. Tests for fat, solids-not-fat and adulterants may also be performed.
5. Acceptance or rejection: Segregate milk that fails the prescribed quality standards.
6. Unloading: Transfer accepted milk to the receiving tank through suitable hygienic equipment, such as a milk can tipper or transfer arrangement.
7. Filtration or clarification: Remove visible foreign matter using appropriate filtration or clarification equipment.
8. Measurement: Measure the quantity of milk received by weighing or calibrated volume measurement.
9. Transfer for processing: Send the milk to the appropriate storage or chilling system.
10. Cleaning: Empty cans are washed, sanitized and prepared for reuse.

B. Unloading milk received in tankers

1. Vehicle inspection: Check tanker identification, seals, cleanliness and delivery documents.
2. Sampling: Collect a representative sample using a hygienic sampling valve or suitable sampling device.
3. Quality testing: Assess temperature, organoleptic characteristics, acidity, fat, solids-not-fat and other specified quality parameters.
4. Acceptance: Confirm that the milk meets the dairy's acceptance criteria before unloading.
5. Connection: Connect the tanker outlet to the dairy's receiving pipeline using clean and sanitized hoses and fittings.
6. Unloading: Operate the transfer pump and discharge milk into the designated receiving tank.
7. Filtration: Pass milk through the specified filter or clarifier, where provided.
8. Quantity measurement: Record the quantity received using a calibrated flow meter or weighing system.
9. Documentation: Record supplier details, quantity, temperature, test results and acceptance status.
10. Post-unloading cleaning: Disconnect the tanker hygienically and clean the tanker and associated equipment according to the prescribed cleaning procedure.

Precautions

- Prevent contamination from dust, insects, dirty equipment and personnel.
- Keep the reception area clean and well organized.
- Minimize delays and unnecessary exposure to warm conditions.
- Never mix rejected milk with accepted milk.
- Follow the dairy's hygiene, sampling and quality-control procedures.

Conclusion: Proper reception, sampling, testing, measurement and hygienic unloading ensure that only acceptable-quality milk enters the processing system.

19. Describe the manufacturing process of yoghurt. Explain with the help of a flow diagram.

Yoghurt is a fermented milk product produced by the action of specific lactic acid bacteria on milk. The characteristic flavour and texture develop through acid production and changes in milk proteins.

Starter cultures

The traditional yoghurt starter consists of:

- Streptococcus thermophilus
- Lactobacillus delbrueckii subsp. bulgaricus

These organisms act symbiotically and convert lactose into lactic acid.

Manufacturing process

Selection and filtration of milk
Standardization of fat and solids
Homogenization
Heat treatment (e.g. 85°C for 30 min or an equivalent validated treatment)
Cooling to inoculation temperature (usually 42–45°C)
Addition of yoghurt starter culture
Incubation and fermentation
Cooling
Packaging (for stirred yoghurt, stirring and filling are performed as appropriate)
Refrigerated storage and distribution

Steps involved

1. Selection and filtration: Fresh, good-quality milk is selected and filtered to remove foreign matter.

2. Standardization: Fat and milk solids are adjusted to obtain the desired composition and consistency.

3. Homogenization: Milk is homogenized to reduce fat globule size and improve the uniformity and stability of the product.

4. Heat treatment: Milk is heated to destroy undesirable microorganisms and denature whey proteins, improving yoghurt firmness and reducing whey separation.

5. Cooling: The milk is cooled to a temperature suitable for the starter culture, generally around 42–45°C for conventional yoghurt cultures.

6. Inoculation: A suitable quantity of active yoghurt starter culture is added under hygienic conditions.

7. Incubation: The inoculated milk is maintained at the appropriate temperature until sufficient acid develops and the desired coagulum forms. The endpoint is commonly around pH 4.5, depending on the product specification.

8. Cooling: Fermentation is stopped or slowed by rapid cooling. For set yoghurt, the product is generally incubated in its final container. For stirred yoghurt, the coagulum is stirred before or during filling, according to the process.

9. Packaging: The yoghurt is packed in clean, food-grade containers that protect it from contamination.

10. Storage: The product is maintained under refrigeration to preserve quality and slow further acid development.

Quality characteristics of good yoghurt

- Smooth, uniform consistency.
- Pleasant, characteristic acidic flavour.
- Absence of objectionable odours.
- Minimal whey separation.
- Appropriate acidity and viable starter organisms.

Conclusion: Yoghurt manufacture depends on proper milk standardization, heat treatment, starter culture activity, controlled fermentation and refrigerated storage.

20. Write an essay on the separation methods and processing of milk cream.

Cream is the fat-rich portion of milk obtained by separating milk fat from the skim milk. It is used in butter, ice cream, bakery products, confectionery and other dairy foods.

A. Methods of cream separation

1. Gravity separation

- Based on the difference in density between milk fat globules and the surrounding milk serum.
- Fat globules rise gradually to the surface when milk is allowed to stand.
- The cream layer is collected from the top.
- This method is simple but slow and less efficient than mechanical separation.

2. Centrifugal separation

- Uses centrifugal force to separate fat globules from the skim milk.
- Milk is fed into a centrifugal cream separator containing a rotating bowl and a series of discs.
- The denser skim milk moves outward, while the lighter fat-rich cream moves towards the centre.
- Cream and skim milk are discharged separately.
- This method is rapid, efficient and widely used in modern dairy plants.

3. Separation by membrane technology

- Certain membrane processes can separate or concentrate milk fat and other milk components.
- The separation achieved depends on the membrane pore size, operating conditions and the particular process.
- Membrane processing is useful for selected dairy applications but is not a universal replacement for centrifugal cream separation.

B. Processing of milk cream

1. Reception and quality testing: Milk is tested for freshness, acidity, fat content and other relevant quality parameters.

2. Filtration or clarification: Foreign matter and suspended impurities are removed.

3. Preheating: Milk may be warmed to a suitable temperature to facilitate efficient centrifugal separation and reduce viscosity.

4. Separation: Milk is passed through a cream separator to obtain cream and skim milk.

5. Standardization: The cream is adjusted to the required fat percentage according to its intended use.

6. Pasteurization: Cream is heat-treated to destroy pathogenic microorganisms and reduce spoilage organisms. The treatment depends on the cream type and intended application.

7. Cooling: Pasteurized cream is cooled to a suitable temperature.

8. Ageing, where required: Cream intended for churning may be cooled and held under controlled conditions to promote the desired crystallization of milk fat.

9. Packaging: Cream intended for sale is filled into clean, suitable packaging under hygienic conditions.

10. Storage and distribution: Cream is stored at the appropriate temperature to maintain its flavour, texture and microbiological quality.

C. Factors affecting cream separation

- Size of fat globules.
- Temperature of milk during separation.
- Speed and efficiency of the separator.
- Fat content of the incoming milk.
- Milk viscosity.
- Proper maintenance and cleanliness of the separator.

D. Uses of cream

- Manufacture of butter and ghee.
- Production of ice cream.
- Preparation of whipped cream.
- Use in bakery and confectionery products.
- Preparation of sauces, desserts and other foods.